Plant Disease Detection

EDA

In [1]:
```
# import Libraries

import os
import matplotlib.pyplot as plt
from PIL import Image
import random
import tensorflow as tf
from tensorflow.keras import models, layers
import matplotlib.pyplot as plt
from IPython.display import HTML
```

In [2]:
```
# Load dataset

data = "data"

# Cheking how big dataset is

total_images = 0

for folder in os.listdir(data):
    folder_path = os.path.join(data, folder)

    if os.path.isdir(folder_path):
        total_images += len(os.listdir(folder_path))

print("Total number of images:", total_images)

# List Clases

classes = os.listdir(data)
print("Total number of classes:", len(classes))
print("class names:", classes[:10])
```

```
Total number of images: 54305
Total number of classes: 38
class names: ['Apple___Apple_scab', 'Apple___Black_rot', 'Apple___Cedar_apple_rust', 'Apple___healthy', 'Blueber
ry___healthy', 'Cherry_(including_sour)___healthy', 'Cherry_(including_sour)___Powdery_mildew', 'Corn_(maize)___
Cercospora_leaf_spot Gray_leaf_spot', 'Corn_(maize)___Common_rust_', 'Corn_(maize)___healthy']
```

In [3]:
```
# Count Images in Each Class

image_count = {}

for cls in classes:
    class_path = os.path.join(data, cls)
    image_count[cls] = len(os.listdir(class_path))

print("Total Images per Class:",image_count)

# Cheking which class has lowest images which has highest images

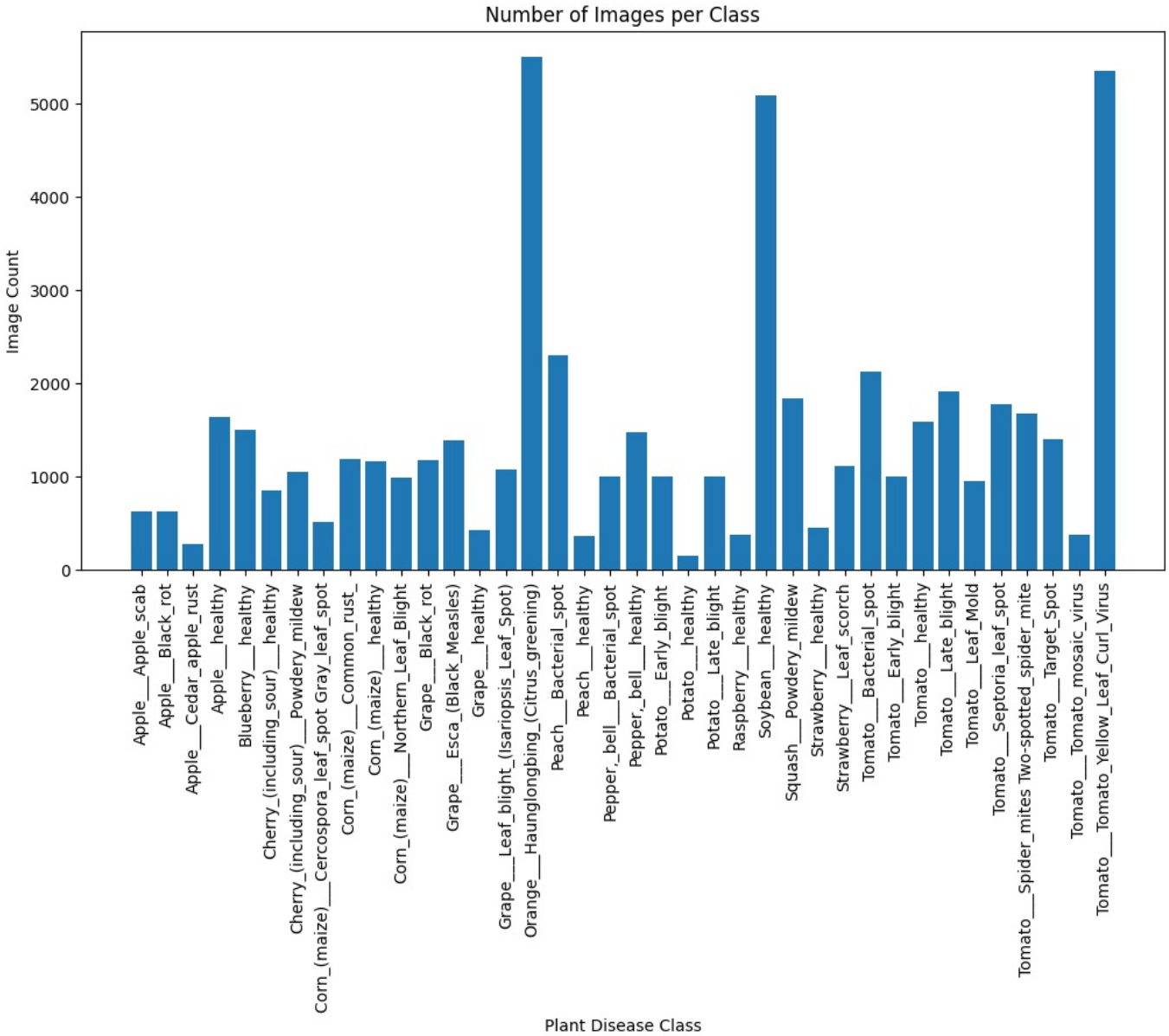
class_counts = {
    cls: len(os.listdir(os.path.join(data, cls)))
    for cls in classes
}

print('Min images in a class:', min(class_counts.values()))
print('Max images in a class:', max(class_counts.values()))
```

```
Total Images per Class: {'Apple___Apple_scab': 630, 'Apple___Black_rot': 621, 'Apple___Cedar_apple_rust': 275, '
Apple___healthy': 1645, 'Blueberry___healthy': 1502, 'Cherry_(including_sour)___healthy': 854, 'Cherry_(includin
g_sour)___Powdery_mildew': 1052, 'Corn_(maize)___Cercospora_leaf_spot Gray_leaf_spot': 513, 'Corn_(maize)___Comm
on_rust_': 1192, 'Corn_(maize)___healthy': 1162, 'Corn_(maize)___Northern_Leaf_Blight': 985, 'Grape___Black_rot'
: 1180, 'Grape___Esca_(Black_Measles)': 1383, 'Grape___healthy': 423, 'Grape___Leaf_blight_(Isariopsis_Leaf_Spot
)': 1076, 'Orange___Haunglongbing_(Citrus_greening)': 5507, 'Peach___Bacterial_spot': 2297, 'Peach___healthy': 3
60, 'Pepper,_bell___Bacterial_spot': 997, 'Pepper,_bell___healthy': 1478, 'Potato___Early_blight': 1000, 'Potato
___healthy': 152, 'Potato___Late_blight': 1000, 'Raspberry___healthy': 371, 'Soybean___healthy': 5090, 'Squash__
_Powdery_mildew': 1835, 'Strawberry___healthy': 456, 'Strawberry___Leaf_scorch': 1109, 'Tomato___Bacterial_spot'
: 2127, 'Tomato___Early_blight': 1000, 'Tomato___healthy': 1591, 'Tomato___Late_blight': 1909, 'Tomato___Leaf_Mo
ld': 952, 'Tomato___Septoria_leaf_spot': 1771, 'Tomato___Spider_mites Two-spotted_spider_mite': 1676, 'Tomato___
Target_Spot': 1404, 'Tomato___Tomato_mosaic_virus': 373, 'Tomato___Tomato_Yellow_Leaf_Curl_Virus': 5357}
Min images in a class: 152
Max images in a class: 5507
```

In [4]:
```
#Plot Class Distribution (Most Important EDA)
```

```
plt.figure(figsize=(12,6))
plt.bar(image_count.keys(), image_count.values())
plt.xticks(rotation=90)
plt.title("Number of Images per Class")
plt.xlabel("Plant Disease Class")
plt.ylabel("Image Count")
plt.show()
```



In [5]:

```
# Show Sample Images (Visual Inspection)

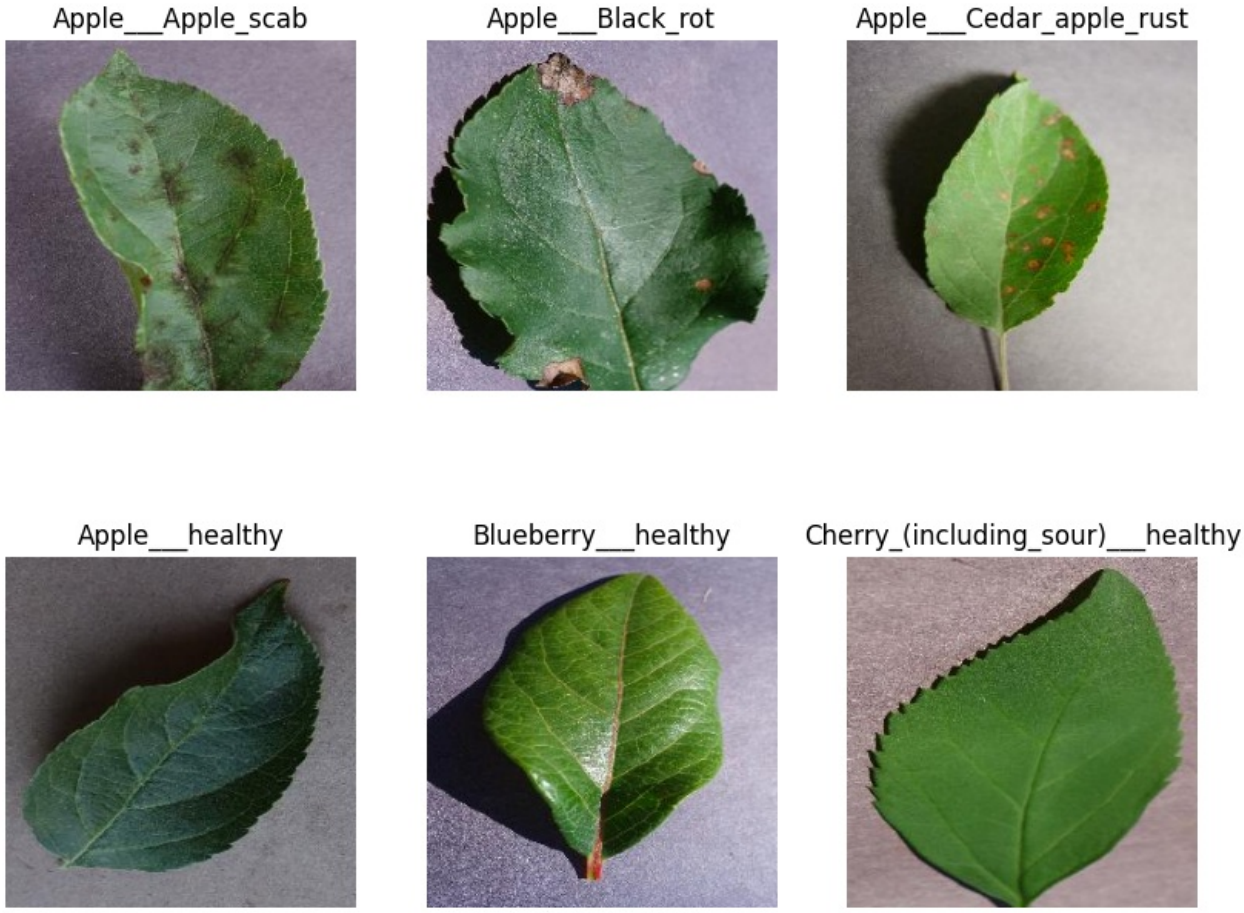
plt.figure(figsize=(10,8))

for i, cls in enumerate(classes[:6]):
    class_path = os.path.join(data, cls)
    img_name = random.choice(os.listdir(class_path))
    img_path = os.path.join(class_path, img_name)

    img = Image.open(img_path)

    plt.subplot(2, 3, i+1)
    plt.imshow(img)
    plt.title(cls)
    plt.axis("off")

plt.show()
```



In [6]:
```
# Check Image Sizes

widths = []
heights = []

for cls in classes[:5]:  # only first 5 classes to keep it simple
    class_path = os.path.join(data, cls)
    for img_name in os.listdir(class_path):
        img = Image.open(os.path.join(class_path, img_name))
        w, h = img.size
        widths.append(w)
        heights.append(h)

print(widths)
print(heights)
```

```
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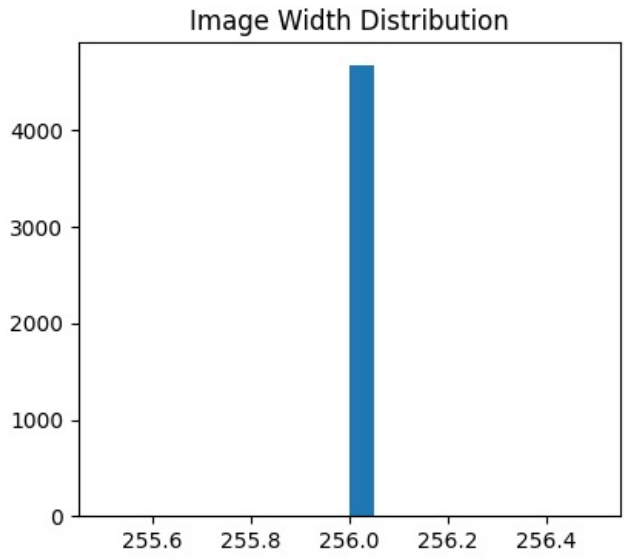
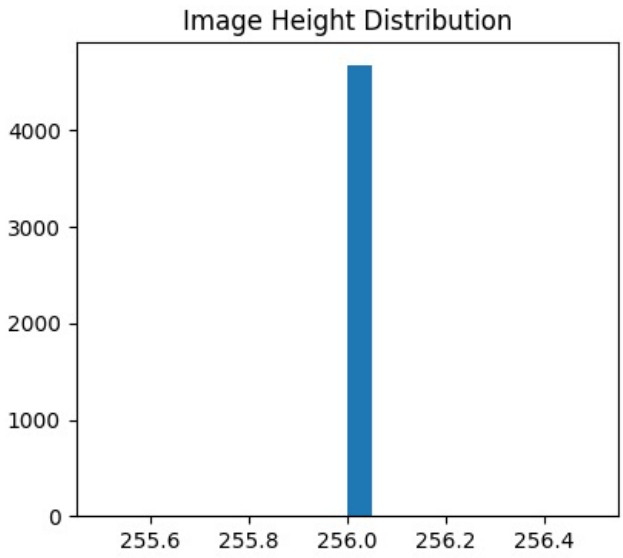


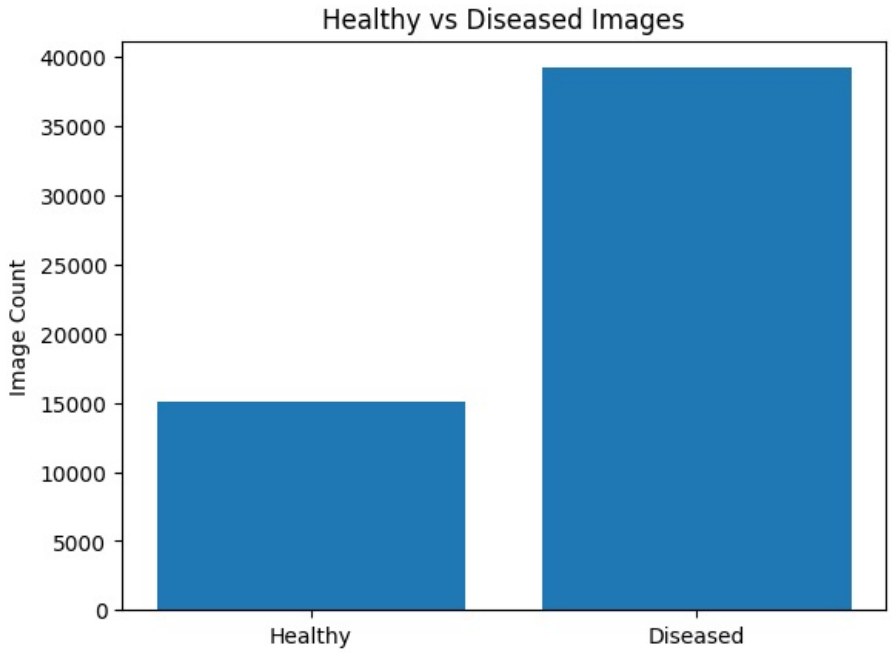


In [8]:
```
# Healthy vs Diseased Images Count

healthy = 0
diseased = 0

for cls in classes:
    if "healthy" in cls.lower():
        healthy += len(os.listdir(os.path.join(data, cls)))
    else:
        diseased += len(os.listdir(os.path.join(data, cls)))

plt.bar(["Healthy", "Diseased"], [healthy, diseased])
plt.title("Healthy vs Diseased Images")
plt.ylabel("Image Count")
plt.show()
```



Model Building

In [9]:
```
# Setting Hyperparameters

BATCH_SIZE = 32
IMAGE_SIZE = 256
CHANNELS=3
EPOCHS=30
```

In [10]:
```
dataset = tf.keras.preprocessing.image_dataset_from_directory(
    "data",
    seed=123,
    shuffle=True,
    image_size=(IMAGE_SIZE,IMAGE_SIZE),
    batch_size=BATCH_SIZE
)
```

```
Found 54305 files belonging to 38 classes.
```

In [11]:
```
len(dataset)
```

Out[11]:
```
1698
```

In [12]:
```
# Making the partion of data into train, validation and test.

def get_dataset_partitions_tf(ds, train_split=0.8, val_split=0.1, test_split=0.1, shuffle=True, shuffle_size=10
    assert (train_split + test_split + val_split) == 1

    ds_size = len(ds)

    if shuffle:
        ds = ds.shuffle(shuffle_size, seed=12)

    train_size = int(train_split * ds_size)
    val_size = int(val_split * ds_size)

    train_ds = ds.take(train_size)
    val_ds = ds.skip(train_size).take(val_size)
    test_ds = ds.skip(train_size).skip(val_size)

    return train_ds, val_ds, test_ds

train_ds, val_ds, test_ds = get_dataset_partitions_tf(dataset)
```

In [13]:
```
#  printing the length of data in particular part

print("Length of train data:",len(train_ds))
print("Length of Validation data:",len(val_ds))
print("Length of test data:",len(test_ds))
```

```
Length of train data: 1358
Length of Validation data: 169
Length of test data: 171
```

In [14]:
```
# Here we are using cache and prefetch to reduce the time in training

# cache is used to store the images in CPU.So, they do not have to look it every time in each epoch and they th
# prefetch is loading the next batch in CPU while running the previous one in GPU.

train_ds = train_ds.cache().shuffle(1000).prefetch(buffer_size=tf.data.AUTOTUNE)
val_ds = val_ds.cache().shuffle(1000).prefetch(buffer_size=tf.data.AUTOTUNE)
test_ds = test_ds.cache().shuffle(1000).prefetch(buffer_size=tf.data.AUTOTUNE)
```

In [15]:
```
# Normaliztion of data
resize_and_rescale = tf.keras.Sequential([
  layers.experimental.preprocessing.Resizing(IMAGE_SIZE, IMAGE_SIZE),
  layers.experimental.preprocessing.Rescaling(1./255),
])

data_augmentation = tf.keras.Sequential([
  layers.experimental.preprocessing.RandomFlip("horizontal_and_vertical"),
  layers.experimental.preprocessing.RandomRotation(0.2),
])
```

In [16]:
```
input_shape = (BATCH_SIZE, IMAGE_SIZE, IMAGE_SIZE, CHANNELS)
n_classes = 33

model = models.Sequential([
    resize_and_rescale,data_augmentation,
    layers.Conv2D(64, kernel_size = (3,3), activation='relu', input_shape=input_shape),
    layers.MaxPooling2D((2, 2)),
    layers.Conv2D(64,  kernel_size = (3,3), activation='relu'),
    layers.MaxPooling2D((2, 2)),
    layers.Conv2D(64,  kernel_size = (3,3), activation='relu'),
    layers.MaxPooling2D((2, 2)),
    layers.Conv2D(64, (3, 3), activation='relu'),
    layers.MaxPooling2D((2, 2)),
    layers.Conv2D(64, (3, 3), activation='relu'),
    layers.MaxPooling2D((2, 2)),
    layers.Conv2D(64, (3, 3), activation='relu'),
    layers.MaxPooling2D((2, 2)),
    layers.Flatten(),
    layers.Dense(64, activation='relu'),
    layers.Dense(n_classes, activation='softmax'),
])

model.build(input_shape=input_shape)
```

In [17]:
```
model.summary()
```

```
Model: "sequential_2"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 sequential (Sequential)     (32, 256, 256, 3)         0

 sequential_1 (Sequential)   (32, 256, 256, 3)         0

 conv2d (Conv2D)             (32, 254, 254, 64)        1792

 max_pooling2d (MaxPooling2D  (32, 127, 127, 64)       0
 )

 conv2d_1 (Conv2D)           (32, 125, 125, 64)        36928

 max_pooling2d_1 (MaxPooling  (32, 62, 62, 64)         0
 2D)

 conv2d_2 (Conv2D)           (32, 60, 60, 64)          36928

 max_pooling2d_2 (MaxPooling  (32, 30, 30, 64)         0
 2D)

 conv2d_3 (Conv2D)           (32, 28, 28, 64)          36928

 max_pooling2d_3 (MaxPooling  (32, 14, 14, 64)         0
 2D)

 conv2d_4 (Conv2D)           (32, 12, 12, 64)          36928

 max_pooling2d_4 (MaxPooling  (32, 6, 6, 64)           0
 2D)

 conv2d_5 (Conv2D)           (32, 4, 4, 64)            36928

 max_pooling2d_5 (MaxPooling  (32, 2, 2, 64)           0
 2D)

 flatten (Flatten)           (32, 256)                 0

 dense (Dense)               (32, 64)                  16448

 dense_1 (Dense)             (32, 33)                  2145

=================================================================
Total params: 205,025
Trainable params: 205,025
Non-trainable params: 0
_________________________________________________________________
```

In [ ]: